Part A:

1. With the complete binary tree, level d will start at 2^d-1 because the initial levels have the nodes of 1+2+…+2^d-1. If the node is at the position j, the index is i=2^d-1+j and its children are at the positions of 2j and 2j+1 at the next level. So, for example: the left would be left=2^d+1-1+2j=2i+1 and the right would be right=2i+2. If you reverse these expressions, you would get the parent index, which would be parent=[(i-1) divided by (2)]. The root has no parent. The child only will exist if only the index is in the list. If insert is at index 13, the sift up path is 13 to 6 to 2 to 0. In this case, the value that is inserted is compared to the indexes of 6,2, and 0. It swaps every time it approaches the root.
2. A completed binary tree with n >1 nodes has a height of log2n. This counts the height in edges. The base case would be a tree with one node which has a height of 0. log2n equals 0, so the formula is good. The inductive hypothesis assumes that the completed binary tree with n nodes has a height of log2n. This means that 2^h is less than or equal to n which is greater than 2 to the h+1. The inductive step adds one node to the next available position. It also checks the tree's height when add another node. So, if the bottom level has room, then the new node would go there, and the height would keep being h. If every level is full, the tree has 2^h+1-1 nodes. When adding another one, it starts another level. Therefore, the height will become h+1. So, by induction, the height is [log2n] for n is greater than or equal to 1.
3.

Insert 5: [5]
Insert 3: [5.3] -> [3,5]
Insert 8: [3,5,8] #No swap
Insert 1: [3,5,8, 1] ->[3,1,8,5] ->[1,3,8,5]
Insert 9: [1,3,8,5,9] #No swap
Insert 2: [1,3,8,5,9,2] ->[1,3,2,5,9,8]
For extract_min(): [8,3,2,5,9] #remove and then return 1, move the last value,8, to root. Swap with the smaller child 2

->[2,3,8,5,9]
The final array is [2,3,8,5,9]

4. If the heap starts at 5, then the doctor would save 5 as the minimum and would pop the last element. So, then, the heap would be empty, and the doctor pauses. Station A inserts 2, which produces [2]. Station B inserts 3, making it be [2,3]. Then, the doctor continues to keep going, sees the nonempty heap, and write the last element of 5 into the root. The is overrides the patient of 2, making it [5,3]. The sift-down will produce [3,5], then the doctor will return 5. Patient 2 is gone, and patient 5 will remain, even with the doctor extracting it. This problem can be prevented if all the stations(A and B) use the exact same lock around the entire insertion or extraction. This will include the sift operation.